CALCULUS · UNIT 7: APPLICATIONS OF INTEGRALS

Glass Vase Studio

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design glass products. The studio logo is an area between curves, vases are volumes of revolution (disc and washer), paperweights use cross sections, and the production rate is integrated for totals.

Pricing glass by volume connects every integral to a cost.

Standards Addressed

- AP CHA-5 Area between curves (x and y)
- AP CHA-5 Volumes with cross sections
- AP CHA-5 Disc and washer methods (about axes and other lines)
- AP CHA-4 Average value and net change

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with photos of blown glass. Part 1 areas.
Day 2	Part 2 vase volumes.
Day 3	Part 3 cross sections, production, and portfolio.

Materials

- Student handout and Specification Cards
- Graphing calculators
- Clay or play-dough to model cross sections (optional)

How the Three Versions Differ

Level	What changes
CP2	Area between two curves, disc method for one vase, and total production.
CP1	Integrating in y, labels, the trig-profile vase, washers, and square cross sections.
Honors	Multi-region areas, equal-area split, rotation about y = −1, wall-thickness optimization, and comparing cross sections.

Watch For (Common Misconceptions)

- Subtracting radii before squaring in the washer method.
- Using the wrong top and bottom curves.
- Forgetting π for discs or the $\frac{\pi}{8}$ factor for semicircles.

Extension Ideas

- 3D-print a vase from your profile.
- Measure a real bottle and model it.
- Compare with the shell method.

Answer Key – CP2

Part 1: Studio Logo

#	Task	Answer
1	Find where the curves intersect.	$\sqrt{x} = \frac{x}{2} \rightarrow x = 0$ and $x = 4$
2	Which curve is on top between $x = 0$ and $x = 4$?	$y = \sqrt{x}$ (e.g., at $x = 1$: $1 > 0.5$)
3	Find the area of the logo.	$\int_0^4 (\sqrt{x} - \frac{x}{2})\,dx = \frac{16}{3} - 4 = \frac{4}{3}$ cm²
4	The logo is etched at \$1.50 per cm². Find the cost.	$\frac{4}{3} \times 1.50 = \$2.00$

Part 2: Bud Vase

#	Task	Answer
5	Write the volume integral using the disc method.	$V = \pi\int_0^{16} (\sqrt{x})^2\,dx = \pi\int_0^{16} x\,dx$
6	Evaluate the volume.	$128\pi \approx 402.1$ cm³
7	Find the cost of the glass at \$0.08 per cm³ (solid paperweight version).	≈ \$32.17
8	Why is $r(x) = \sqrt{x}$ the radius?	It is the distance from the axis (x-axis) to the curve.

Part 3: Production

#	Task	Answer
9	Find the total number of vases made in 4 hours.	$\int_0^4 (6 + 2t)\,dt = 24 + 16 = 40$ vases
10	Find the average production rate.	$\frac{40}{4} = 10$ vases per hour
11	Design a logo with two curves of your own and find its area.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why do you subtract the bottom curve from the top curve?	The height of each thin rectangle is the top minus the bottom.
R2	Why does the disc method square the radius?	Each slice is a circle with area πr^2.

Answer Key – CP1

Part 1: Logo & Labels

#	Task	Answer
1	Find the area of the logo between $y = \sqrt{x}$ and $y = \frac{x}{2}$.	$\frac{4}{3}$ cm²
2	Find the area of the same region by integrating with respect to y.	x from y^2 to 2y: $\int_0^2 (2y - y^2)\,dy = 4 - \frac{8}{3} = \frac{4}{3}$ ✓
3	A label is the region between $y = 4 - x^2$ and $y = x + 2$. Find its area.	Intersections x = −2, 1: $\int$ from −2 to 1 of $(2 - x - x^2)\,dx = \frac{9}{2}$ cm²
4	Explain when integrating in y is easier than in x.	When the region is bounded on the left and right by single curves.

Part 2: Vases

#	Task	Answer
5	Find the volume of the solid tall vase $r = 4 + \sin(\frac{\pi x}{10})$, $0 \le x \le 20$.	$\pi\int_0^{20} (16 + 8\sin(\frac{\pi x}{10}) + \sin^2(\frac{\pi x}{10}))\,dx = \pi(320 + 0 + 10) = 330\pi \approx 1{,}036.7$ cm³
6	Find the volume of glass in the hollow vase (washer method).	$\pi\int_0^{20} [(4 + s)^2 - (3.5 + s)^2]\,dx = \pi\int_0^{20} (3.75 + s)\,dx = \pi(75 + 0) = 75\pi \approx 235.6$ cm³ ($s = \sin(\frac{\pi x}{10})$)
7	Find the glass cost of the hollow vase and the volume of water it holds.	\$18.85; water inside: $\pi\int_0^{20} (3.5 + s)^2\,dx = \pi(245 + 10) = 255\pi \approx 801.1$ cm³
8	Find the volume of the bud vase $y = \sqrt{x}$ rotated about the x-axis for $0 \le x \le 16$.	$128\pi \approx 402.1$ cm³

Part 3: Paperweight & Production

#	Task	Answer
9	Find the volume of the paperweight.	$\int$ from −2 to 2 of $(4 - x^2)^2\,dx = \frac{512}{15} \approx 34.1$ cm³
10	Find the glass cost.	≈ \$2.73
11	Production rate $p(t) = 6 + 2t$ for $0 \le t \le 4$. Find total vases and price the day's glass if each is a hollow vase.	40 vases × \$18.85 ≈ \$754

Reflection

#	Prompt	Look for
R1	How is the washer method related to the disc method?	It subtracts the inner disc volume from the outer disc volume.
R2	How do cross sections generalize volume?	Integrate the area of any cross-section shape along the axis.

Answer Key – Honors

Part 1: Logo Design

#	Task	Answer
1	Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ and the line $x = 9$ (two regions).	$\int_0^4 (\sqrt{x} - \frac{x}{2})\,dx + \int_4^9 (\frac{x}{2} - \sqrt{x})\,dx = \frac{4}{3} + (\frac{65}{4} - \frac{38}{3}) = \frac{4}{3} + \frac{43}{12} = \frac{59}{12}$ cm²
2	Find k so the vertical line $x = k$ splits the logo ($0 \le x \le 4$) into two equal areas.	$\frac{2}{3}k\sqrt{k} - \frac{k^2}{4} = \frac{2}{3} \rightarrow k \approx 1.51$ (technology)
3	Rotate the logo about the x-axis. Find the volume (washer method).	$\pi\int_0^4 (x - \frac{x^2}{4})\,dx = \pi(8 - \frac{16}{3}) = \frac{8\pi}{3} \approx 8.38$ cm³
4	Rotate the logo about the line $y = -1$. Find the volume.	$\pi\int_0^4 [(\sqrt{x}+1)^2 - (\frac{x}{2}+1)^2]\,dx = \pi\int_0^4 (x + 2\sqrt{x} - \frac{x^2}{4} - x)\,dx = \pi(\frac{32}{3} - \frac{16}{3}) = \frac{16\pi}{3} \approx 16.76$ cm³

Part 2: Vase Engineering

#	Task	Answer
5	Find the glass volume and the inside (water) volume.	Glass $75\pi \approx 235.6$ cm³; water $255\pi \approx 801.1$ cm³
6	The wall thickness is changed to w (inner $r = 4 - w + s$). Write the glass volume as a function of w.	$V(w) = \pi\int_0^{20} [2w(4+s) - w^2]\,dx = \pi(160w - 20w^2)$
7	The vase must hold at least 850 cm³ of water. What is the greatest wall thickness allowed?	Water $\pi\int (4 - w + s)^2\,dx = \pi(20(4-w)^2 + 10) \ge 850 \rightarrow (4-w)^2 \ge 13.03 \rightarrow w \le 0.39$ cm
8	Find the glass cost for that thickness.	$V(0.39) \approx \pi(62.4 - 3.04) \approx 186.7$ cm³ → ≈ $14.94

Part 3: Paperweights

#	Task	Answer
9	Compare the volumes with square and semicircular cross sections perpendicular to the x-axis.	Square $\frac{512}{15} \approx 34.1$; semicircle (diameter $4 - x^2$): $\frac{\pi}{8}\int (4 - x^2)^2\,dx = \frac{\pi}{8} \cdot \frac{512}{15} \approx 13.4$ cm³
10	Rotate the base about the y-axis. Use the disc method in y to find the volume.	$\pi\int_0^4 (4 - y)\,dy = 8\pi \approx 25.1$ cm³
11	Write the design portfolio with three products, their integrals, volumes, and prices.	Answers vary.

Reflection

#	Prompt	Look for
R1	How does rotating about a different axis change the radius?	The radius is the distance from the axis, so you add or subtract the axis position.
R2	Why did the water-volume requirement limit wall thickness?	Thicker walls shrink the inner radius and the volume it holds.